

M3_Assignment

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Q2 The AMA

The American Medical Association must determine how many rooms to reserve six months before its annual convention. Currently, the AMA can reserve rooms at \$150 per room. The AMA believes the number of doctors attending the convention will be normally distributed with a mean of 5000 and a standard deviation of 1000. If the number of people attending the convention exceeds the number of rooms reserved, extra rooms must be reserved at \$250 per room.

Q2.1 Basic Model

Write code below that captures the inputs, decision variables and random variables of the model. Name the object for the decision variable `num_reservations` and set its value to 5,000. The inputs should be called `regular_cost`, and `late_cost` (both are per room). Finally, the random variable should be called `attendance` and 10,000 numbers should be generated from the normal distribution with the seed set at 10.

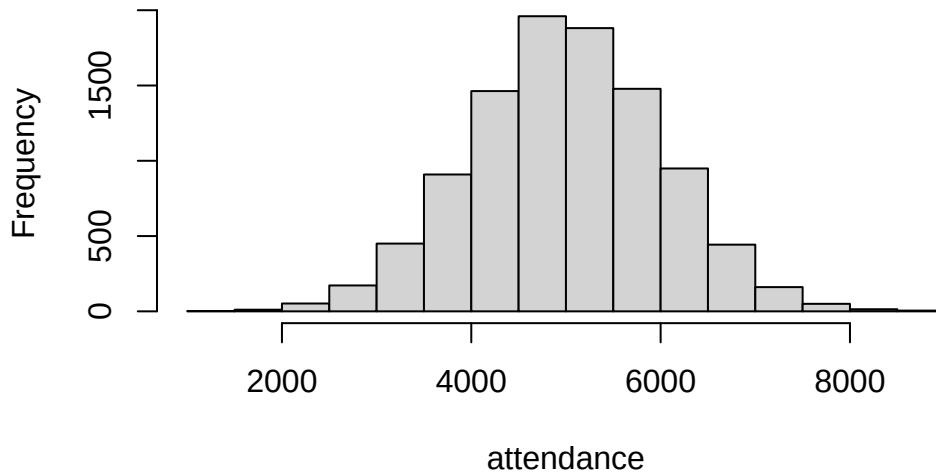
```
#Decision Variable
num_reservations <- 5000

#Inputs
regular_cost <- 150
late_cost <- 250

#Random Variable
set.seed(10)
n <- 10000
attendance <- rnorm(n, 5000, 1000)

hist(attendance)
```

Histogram of attendance



Q2.2 Extra Rooms

Create an object with an empty vector called `extra_rooms` to record the number of extra rooms needed. Use a for loop to populate the vector by comparing the attendance and `num_reservations` objects. This vector should have 10,000 elements. Include the code below:

```
extra_rooms <-c()
for (i in 1:n) {
  extra_rooms <- pmax(0, round(attendance - num_reservations))
}
```

Q2.3 Reservation Cost

Create an object called `reservation_cost` using the `rep()` function with the `times` argument set to 10,000. Given the number of reservations and the regular cost, this vector should capture the 10,000 reservation costs. Include the code below:

```
reservation_cost <- rep(num_reservations*regular_cost, n)
```

Q2.4 Total Cost

Create an object called `late_reservation_cost` that captures the late reservation cost given the number of extra rooms needed for each simulation. Use this object to create a final object that captures the total cost and call it `total_cost`.

```
late_reservation_cost <- extra_rooms*late_cost  
  
total_cost <- reservation_cost + late_reservation_cost
```

Q2.5 Statistics

What is the approximate mean, and standard deviation of the `total_cost` object?

```
mean(total_cost)
```

```
[1] 850344.8
```

```
sd(total_cost)
```

```
[1] 147069.1
```

Q2.6 Optimal Number of Rooms

Use the model developed above to determine the number of rooms that should be reserved to minimize the expected cost to the AMA. Try possible values from 4000 to 5000 in increments of 100. You can use the model built in Q2.1-Q2.5 to answer this question by just changing the `num_reservations` value (no need to use a for loop). How many rooms should be reserved to minimize the average cost?

```
###Did a for loop to verify results.  
# Define a range of reservation levels  
reservations <- seq(4000, 5000, 100)  
  
all_reservation_costs<-c()  
  
for (i in reservations) {  
  n <- 10000  
  v_reservations<-rep(i,n)
```

```

v_regular_cost<-rep(regular_cost,n)
v_late_cost<-rep(late_cost,n)

set.seed(10)
attendance <- rnorm(n, 5000, 1000)

extra_rooms<-pmax(0, attendance - v_reservations)

reservation_cost<-v_reservations*v_regular_cost
late_reservation_cost<-extra_rooms*v_late_cost
total_costs<-reservation_cost+late_reservation_cost

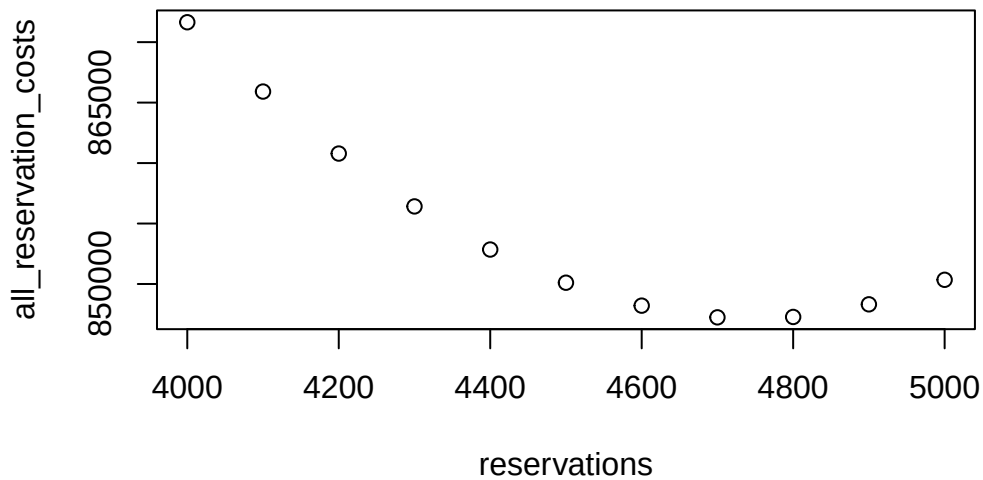
all_reservation_costs<- c(all_reservation_costs,mean(total_costs))
}

results<-data.frame(reservations,all_reservation_costs)
print(results)

```

	reservations	all_reservation_costs
1	4000	871646.9
2	4100	865914.6
3	4200	860787.2
4	4300	856415.6
5	4400	852849.6
6	4500	850104.8
7	4600	848207.7
8	4700	847239.3
9	4800	847272.8
10	4900	848316.3
11	5000	850344.8

```
plot(results)
```



Q2.7 Discussion

Provide your recommendation for the AMA below. Discuss how the mean attendance is a good/bad estimation of how many rooms should be reserved? Explain how your recommendation might be superior than just booking the mean attendance of 5000.

I recommend booking 4700 rooms to minimize expected cost to the AMA. Though mean can be a helpful estimation, it can also be misleading because it does not account for the variability or distribution of attendance. By simulating thousands of scenarios based on a probability distribution, we can factor in uncertainty and better understand the range of outcomes and associated risks.